Phonon Wave Equation in TauTimp Onduscular Framework **Author:** Viorel Popescu **Framework:** TauTimp — Onduscular Theory (Teoria Onduscular) **Date:** 2026-03-30 1. Setup and Wave Equation

The TauTimp onduscular wave equation with variable coefficient, spherical symmetry (f independent of θ, ϕ):

$$\frac{R}{r}\,\Delta f(r,t) = \frac{1}{V^2}\,\frac{\partial^2 f(r,t)}{\partial t^2}$$

with $R = 4.2355 \times 10^{22}$ m, $V = 346$ m/s, and $\Delta f = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}$

Full PDE: $\boxed{\frac{R}{r}\left(\frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}\right) = \frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}}$

2. Separation of Variables

With $f(r,t) = g(r)\,u(t)$ and separation constant $-\lambda^2$:

Temporal part:

$$u''(t) + \lambda^2 V^2\,u(t) = 0 \quad \Rightarrow \quad u(t) = A\cos(\omega t) + B\sin(\omega t), \quad \omega = \lambda V$$

Radial part — substitution $g(r) = h(r)/r$:

$$h''(r) + \frac{\lambda^2\, r}{R}\,h(r) = 0$$

3. Airy Equation and Eigenfunctions

With $\xi = -\left(\frac{\lambda^2}{R}\right)^{1/3} r$ the radial ODE becomes the Airy equation:

$$\frac{d^2 h}{d\xi^2} - \xi\,h(\xi) = 0 \quad \Rightarrow \quad h(\xi) = C_1\,\mathrm{Ai}(\xi) + C_2\,\mathrm{Bi}(\xi)$$

Regularity at the origin requires $C_2 = 0$. The boundary condition $g(R) = 0$ quantizes:

$$\mathrm{Ai}(a_n) = 0 \quad \Rightarrow \quad \boxed{\omega_n = \frac{V\,|a_n|^{3/2}}{R}}$$

First zeros: $a_1 \approx -2.3381$, $a_2 \approx -4.0879$, $a_3 \approx -5.5206$.

Radial eigenfunctions: $g_n(r) = \frac{A_n}{r}\,\mathrm{Ai}\left(\frac{a_n r}{R}\right)$

Full field:

$$f(r,t) = \sum_{n=1}^{\infty} \frac{A_n}{r}\,\mathrm{Ai}\left(\frac{a_n r}{R}\right)\cos(\omega_n t)$$

4. Construction of $H(r,\theta,t)$

$$H(r,\theta,t) = \sum_{\ell=0}^{\infty} f_\ell\,(r,t)\,P_\ell(\cos\theta)$$

For spherically symmetric modes ($\ell = 0$, $P_0 = 1$): $H(r,\theta,t) = f(r,t)$.

5. Probability Density Formalism

Define $\psi(r,t) \equiv f(r,t)$ and:

$$P(r,t) = |\psi(r,t)|^2 = f(r,t)^2$$

P satisfies simultaneously:

Flux wave equation:

$$\frac{1}{V^2}\frac{\partial^2 P}{\partial t^2} - \frac{\partial^2 P}{\partial r^2} = 0$$

Transport (outgoing characteristic) condition:

$$\frac{\partial P}{\partial r} = \frac{1}{V}\frac{\partial P}{\partial t}$$

Together these force $P = P(r - Vt)$, an outgoing wave.

Boundary condition: $P(R,0) = 0$.

6. Degree-2 Polynomial Solution

We seek $P(r,t) = q_0 + q_1(r - Vt) + q_2(r - Vt)^2$, which automatically satisfies both the wave equation and the transport condition since it depends only on $r - Vt$.

6.1 Constraints from the boundary condition

At $t = 0$: $P(R,0) = q_0 + q_1 R + q_2 R^2 = 0$.

We impose additionally $P(0,0) = 0$ (vanishing at the origin) to obtain a physically natural **interior bell shape**. These two roots $r = 0$ and $r = R$ fix the factored form:

$$P^{(2)}(r,0) = q_2\,r\,(r - R)$$

For $P \geq 0$ on $[0,R]$ we need $q_2 < 0$. Normalizing so that $\max P = P_0 = 1$: the maximum occurs at $r = R/2$ and equals $-q_2 R^2/4 = 1$, giving $q_2 = -4/R^2$. Therefore:

$$\boxed{P^{(2)}(r,0) = \frac{4}{R^2}\,r\,(R - r)}$$

6.2 Explicit polynomial coefficients

$$q_0 = 0, \qquad q_1 = \frac{4}{R}, \qquad q_2 = -\frac{4}{R^2}$$

Numerically with $R = 4.2355 \times 10^{22}$ m:

Coefficient	Value
q_0	0
q_1	$9.4439 \times 10^{-23}\ \mathrm{m}^{-1}$
q_2	$-2.2297 \times 10^{-45}\ \mathrm{m}^{-2}$

6.3 Verification of boundary conditions

$$P^{(2)}(0,\,0) = \frac{4}{R^2} \cdot 0 \cdot R = 0 \quad \checkmark$$

$$P^{(2)}(R,\,0) = \frac{4}{R^2} \cdot R \cdot (R - R) = 0 \quad \checkmark$$

$$P^{(2)}\left(\frac{R}{2},\,0\right) = \frac{4}{R^2} \cdot \frac{R}{2} \cdot \frac{R}{2} = 1 = P_0 \quad \checkmark$$

6.4 Full time-dependent solution

Replacing $r \to r - Vt$:

$$\boxed{P^{(2)}(r,t) = \frac{4}{R^2}\,(r - Vt)\,(R - r + Vt)}$$

Check: $P^{(2)}(R,0) = \frac{4}{R^2} \cdot R \cdot 0 = 0$ ✓.

7. Phonon Flux Density (Degree-2)

The flux per unit solid angle area:

$$\mathcal{F}^{(2)}(r,t) = \frac{P^{(2)}(r,t)}{4\pi r^2}$$

At $t = 0$:

$$\mathcal{F}^{(2)}(r,0) = \frac{4}{R^2} \cdot \frac{r\,(R-r)}{4\pi r^2} = \frac{1}{\pi R^2} \cdot \frac{R-r}{r}$$

$$\boxed{\mathcal{F}^{(2)}(r,0) = \frac{1}{\pi R^2} \cdot \frac{R-r}{r}}$$

Numerically:

$$\mathcal{F}^{(2)}(r,0) = \frac{1}{\pi\,(4.2355 \times 10^{22})^2} \cdot \frac{R-r}{r} = 1.774 \times 10^{-46} \cdot \frac{R-r}{r} \quad \mathrm{m}^{-2}$$

Spot values:

r	$\mathcal{F}^{(2)}(r,0)$ [m^{-2}]
$R/4$	5.323×10^{-46}
$R/2$	1.774×10^{-46}
$3R/4$	5.914×10^{-47}

8. Summary of Key Results

Quantity	Expression
Radial eigenfunction	$g_n(r) = (A_n/r)\,\mathrm{Ai}(a_n r/R)$
Eigenfrequency	$\omega_n = V\lvert a_n\rvert^{3/2}/R$
Full wave field	$f(r,t) = \sum_n g_n(r)\cos(\omega_n t)$
Phonon field	$H(r,\theta,t) = \sum_\ell f_\ell\, P_\ell(\cos\theta)$
Prob. density (deg-2)	$P^{(2)}(r,0) = (4/R^2)\,r(R-r)$
Full $P^{(2)}(r,t)$	$(4/R^2)(r - Vt)(R - r + Vt)$
q_0	0
q_1	$4/R = 9.4439 \times 10^{-23}\ \mathrm{m}^{-1}$
q_2	$-4/R^2 = -2.2297 \times 10^{-45}\ \mathrm{m}^{-2}$
Flux $\mathcal{F}^{(2)}(r,0)$	$(1/\pi R^2)(R-r)/r$
R	4.2355×10^{22} m
V	346 m/s

TauTimp Onduscular Theory — Phonon Field Analysis

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